Immigration and Credit in America

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TransUnion (the data provider) has the right to review the research before dissemination to ensure it accurately describes TransUnion data, does not disclose confidential information, and does not contain material it deems to be misleading or false regarding TransUnion, TransUnion's partners, affiliates or customer base, or the consumer lending industry.

Motivation

- Immigrants to the U.S. bring their labor market experiences (e.g., education records, work history, human capital) helping them to navigate U.S. labor markets

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- Immigrants start life in the U.S. *without* a U.S. credit report
  - Any foreign credit history generally unobserved to U.S. lenders

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

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- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant**, & **year of immigration**, using sequential Social Security Number (SSNs) assignment

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What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, conditional on birth year & geographic fixed effects

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- U.S. consumer credit reporting data matched with immigration data

What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
  - Compare Immigrants vs. Non-Immigrants, Immigrants arriving at different ages

Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans **due to auto loan debt aversion**
- Credit card limits are **lower** but become **higher over time**

Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Sequeira, Nunn, & Qian, 2020; Kerr & Kerr, 2020; Azoulay, Jones, Kim, & Miranda, 2022; Bernstein, Diamond, McQuade, Jiranaphawiboon, & Pousada, 2025)

- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1987; Abramitzsky & Boustan, 2017)

- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022; Bailey et al., 2022)

- We contribute to the household finance literatures on economics of credit information, & credit use across the life cycle that largely has not studied role of immigrants (e.g., Pagano & Jappelli, 1996; Dobbie et al., 2021; Blattner, Hartwig, & Nelson, 2022; Zillessen, 2022; Bach et al., 2023; Ricks & Sandler, 2025; Bakker et al., 2025)

Roadmap For Presentation

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

Data & Institutional Details

Classifying Immigrants

Before 2012, U.S. Social Security Administration assigned first 5 digits of Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants
- validated in prior work (Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein, Diamond, McQuade, Jiranaphawiboon, & Pousada, 2025)

- **"Immigrants":** assigned SSN at age 21+. **"Non-Immigrants":** assigned SSN at age < 21

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Classifying Immigrants In The BTCCP

1. Created lookup table: map first 5 digits of SSN to SSN assignment year (*SSN Year*)
2. Sent to TransUnion: (i) lookup table (ii) list of anonymous consumer identifiers in BTCCP
3. TransUnion merged to their non-anonymous data that includes SSNs. TransUnion returned consumer SSN Year (without SSNs) linked to anonymous consumer identifiers
4. We calculate age of SSN assignment: $\textit{SSN Age} = \textit{SSN Year} - \textit{Birth Year}$

Data

- University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)
  - Anonymized data sample of 1 in 10 consumers with U.S. credit reports
  - Monthly, tradeline-level data from 2000 to 2024

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang
(2025, *Journal of Economic Literature*)

Sample Restrictions

- Birth years 1975 - 1987
- SSN Age up to 29
- 6,122,932 consumers, 344,261 (6%) SSN Age 21 to 29 (aligns with official statistics)

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

Institutional Details On Lending In America

- Immigrant credit access may be adversely affected by **information frictions**:
  1. **Not having a credit report on arrival**
     - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
  2. **After entering the credit system, have a shorter history than Non-Immigrants**
     - Length of credit report has positive impact to credit scores ($\approx$15% FICO)

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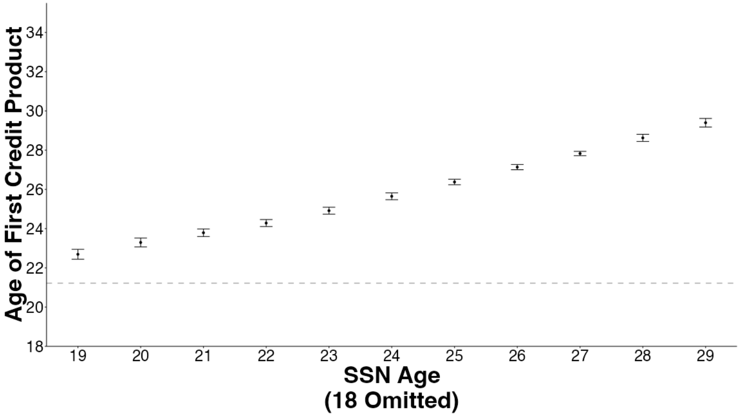
- Different cultural **tastes** & informational barriers for consumers
  - Differences in familiarity & willingness to use debt, understanding of how to build credit score
- Lenders use credit scores (+ other information) in decisions
  - Can use immigration status
  - Cannot use protected characteristics (e.g., gender, national origin, race) under Equal Credit Opportunity Act (ECOA)

Credit Market Entry

SSN Assigned At Older Age → Later Entry Into U.S. Credit System

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \tag{1}$$



Main Cross-Sectional Regression Specification

Cross-Sectional OLS Regression:

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \tag{1}$$

Given **linearity** in *SSN Age*, we mainly use simpler OLS cross-sectional regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \tag{2}$$

β_1: average difference for Immigrants vs. Non-Immigrants

β_2: marginal difference for SSN Age one year later

Cluster standard errors by birth year

Results robust to adding geographic fixed effects to control for local market conditions (First observed ZIP5, Last observed ZIP5, Longest-held ZIP5, number of ZIP5s)

Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

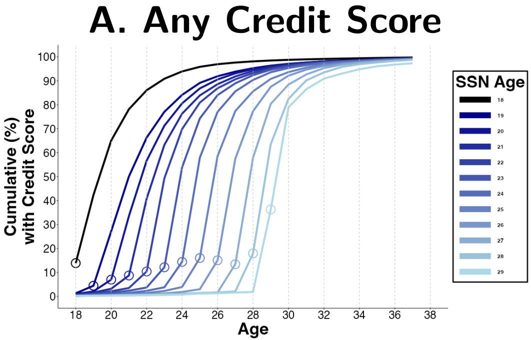
$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \tag{2}$$

Age at First...	Credit Report (1)	Credit Product (2)
21+	1.972***	1.665***
	(0.101)	(0.112)
SSN Age	0.740***	0.704***
	(0.015)	(0.017)
Birth Year F.E.	X	X
R^2	0.267	0.087
Mean, SSN Age <21	19.86	21.25

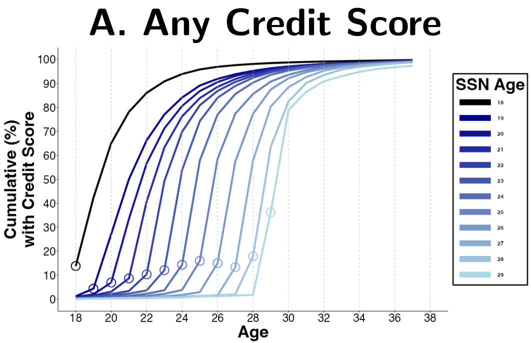
$N = 6,122,932$

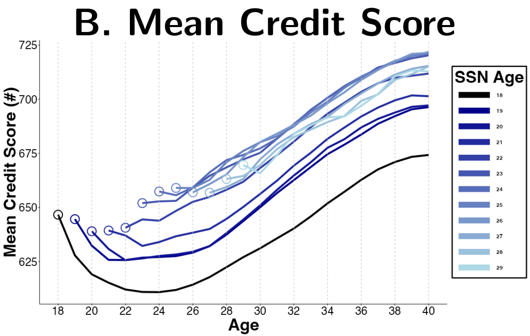
Creditworthiness

Credit Score

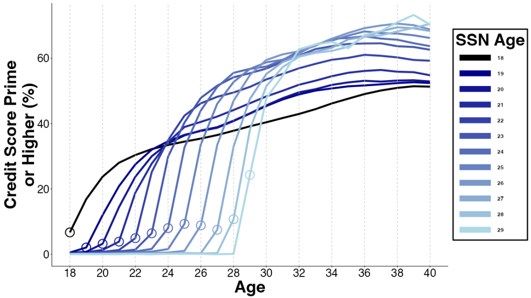


Credit Score





C. Prime or Higher Credit Score (660+)



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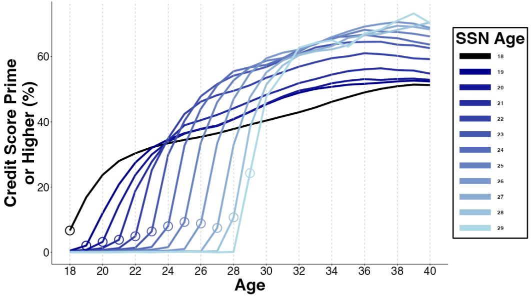
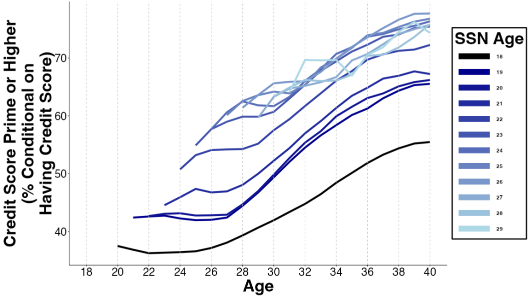


D. Prime or Higher Credit Score (Conditional on Any Score)

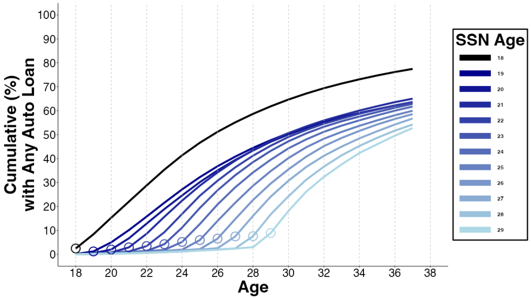


Immigrants (vs. Non-Immigrants) have more compressed credit scores: $\downarrow$ Pr(subprime), $\uparrow$ Pr(superprime)

Credit Access

Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

A. Any Auto Loan



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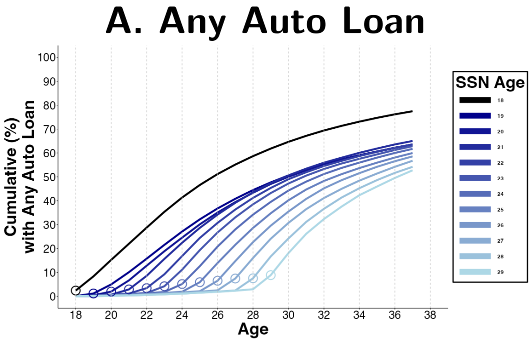

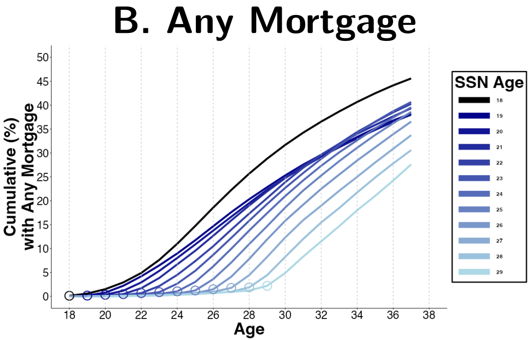

By Age 37, Immigrants are:

10.2 (s.e. 0.4) p.p., 13% *less* likely to have auto loan

- Each year older SSN Age, 1.4 (0.1) p.p. *less* likely to have auto loan

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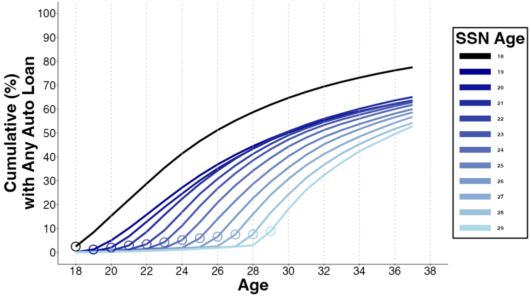
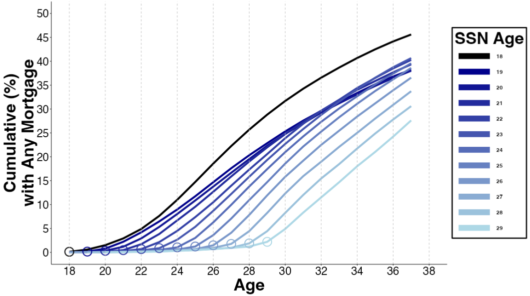


B. Any Mortgage



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Immigrant vs. Non-Immigrant difference in mortgage use *not* robust to geographic F.E.

- Each SSN Age year is 1.6 (0.1) p.p. *less* likely to have mortgage

Immigrants, & Older SSN Age Immigrants, Only First Access Mortgage, Auto Loan, & Credit Card At Older Ages

$$Y_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \tag{2}$$

Age at First...	**Mortgage** (1)	**Auto Loan** (2)	**Credit Card** (3)
21+	0.781^{***}	2.930^{***}	1.160^{***}
	(0.097)	(0.217)	(0.169)
SSN Age	0.403^{***}	0.612^{***}	0.663^{***}
	(0.018)	(0.023)	(0.022)
Birth Year F.E.	X	X	X
R^2	0.024	0.030	0.033
Mean, SSN Age $<$21	36.36	29.15	22.68

$N = 6,122,932$

"Paired Cohorts": Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_\tau \left(\mathbb{I}\{focal\ cohort\}_{k(p)} \times \mathbb{I}\{t=\tau\}\right) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \tag{3}$$

- 'Focal Cohort': $SSN\ Age_{c'}$ 22-30
- 'Control Cohort': $SSN\ Age_{c'} - 1$, same birth year as focal. 68 pairs (p)
- 2,176 cohort-by-year observations ($k(p), t$), s.e. clustered by birth year

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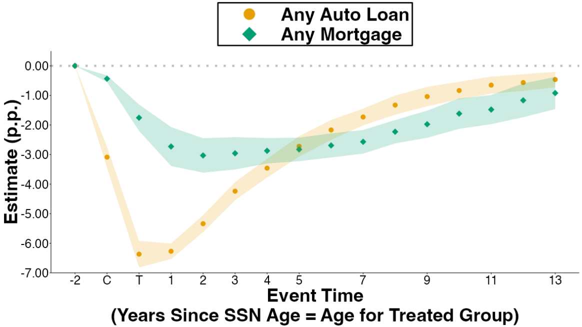
- 2,176 cohort-by-year observations ($k(p), t$), s.e. clustered by birth year
- **Results:**

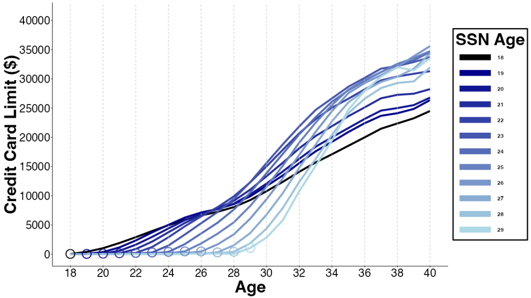
  Any Auto Loan after 10 years:
  -0.46 p.p., -0.42 cumulative years

  Any Mortgage after 10 years:
  -0.92 p.p., -0.29 cumulative years



Immigrants' Credit Card Limits Are Initially Lower Than Non-Immigrants, But Become Higher Than Non-Immigrants Over Life Cycle

A. Credit Card Limits ($)



Similar pattern for credit card limit per card

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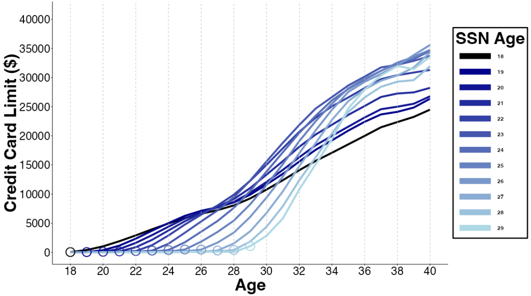

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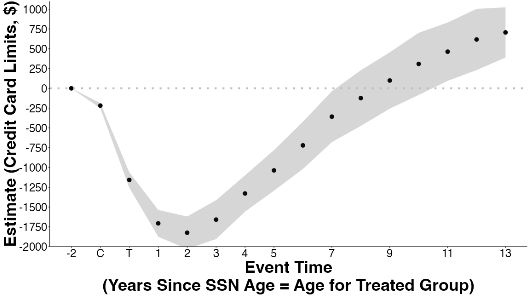
B. Paired Cohorts



SSN Age $+2$ **Years:**

$-\$1,825$ (s.e. $105)

SSN Age $+13$ **Years:**

$+\$707$ (s.e. $162)

Mechanisms

What Mechanisms Explain Our Key Results?

Potential Mechanisms	↓ Immigrant (Vs. Non-Immigrant) Access To... Auto Loans (1)	↓ Access For Immigrants Arriving at Older (Vs. Younger) Ages... Auto Loans (2)	Mortgages (3)	Immigrants Arriving At Older (Vs. Younger) Ages Have... Lower Early Credit Card Limits (4)	Higher Later Credit Card Limits (5)
A. Creditworthiness	✗	✗	✗	✗	✓
B. Emigration	✗	✗	✗	✗	✗
C. Tastes	✓	✓	✓	✓	✓
D. Information Frictions		✓	✓	✓	✓

Information Frictions:

- Immigrants arrive without a U.S. credit report
- Immigrants have a shorter U.S. credit history

Immigrants Are Averse To Purchasing Autos With Debt

Dep Var: Good to Borrow...	In General	For Vacation	For Living Expenses	For Auto Purchase	For Education
	(1)	(2)	(3)	(4)	(5)
21+	-0.1499	-0.0774	-0.0804	-0.2273***	0.0905
	(0.1375)	(0.0518)	(0.0781)	(0.0712)	(0.0559)
Immigration Age	0.0100	0.0041	-0.0011	0.0113*	-0.0149**
	(0.0122)	(0.0056)	(0.0075)	(0.0060)	(0.0068)
F.E. Birth Year	X	X	X	X	X
F.E. Demographic Controls	X	X	X	X	X
R^2	0.030	0.018	0.029	0.072	0.052
N	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	0.0380	0.1530	0.7338	0.7958	0.8424

Immigrants More Likely Than Non-Immigrants To Buy Autos With Cash

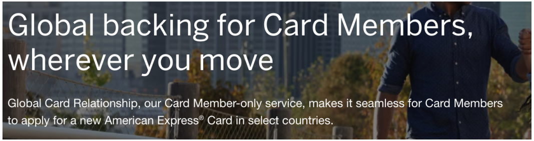
Dep Var:		Conditional On Having An Auto Vehicle			Conditional On Homeowner
	Has Auto Vehicle	Auto Cash	Auto Finance/Lease	Homeowner	No Mortgage
	(1)	(2)	(3)	(4)	(5)
21+	-0.0620	0.1254*	-0.0953	0.0241	0.0704
	(0.0477)	(0.0753)	(0.0921)	(0.0675)	(0.0791)
Immigration Age	-0.0003	-0.0164**	0.0124	-0.0155***	-0.0100*
	(0.0044)	(0.0077)	(0.0077)	(0.0055)	(0.0058)
F.E. Birth Year	X	X	X	X	X
F.E. Demographic Controls	X	X	X	X	X
R^2	0.156	0.038	0.062	0.298	0.175
N	1,813	1,568	1,568	1,813	883
Mean, Immigration Age <21	0.8923	0.7396	0.5283	0.5398	0.1619

Immigrants Have Similar Or Lower Demand for Debt Than Non-Immigrants

Dep Var: Apply for...	Any Credit	Auto	Mortgage	Refinance Mortgage	Credit Card	Limit Increase	Other Credit
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
21+	-0.0071	-0.0690	0.0625	0.0437	0.0519	-0.0279	-0.0729
	(0.0695)	(0.0739)	(0.0700)	(0.0401)	(0.0843)	(0.0609)	(0.0458)
Immigration Age	0.0050	0.0112	-0.0036	-0.0048	-0.0102	0.0073	0.0128**
	(0.0060)	(0.0072)	(0.0067)	(0.0039)	(0.0086)	(0.0062)	(0.0055)
F.E. Birth Year	X	X	X	X	X	X	X
F.E. Demographic Controls	X	X	X	X	X	X	X
R^2	0.059	0.059	0.056	0.033	0.039	0.043	0.040
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	0.6481	0.1995	0.1128	0.0670	0.4043	0.1533	0.1208

Information Friction Of Non-Portability Of Credit Histories Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions:

American Express Global Card Relationship



Nova Credit

International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

Equifax Canada Global Consumer Credit File

Global Consumer Credit File

Helping you grow and retain your clients who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

Conclusions

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection** & **assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants
- exploit variation in age of immigration in credit data matched with immigration information

Immigrants (vs. Non-Immigrants), & Immigrants arriving at older ages:

- **Positively selected**: more creditworthy (once scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans **due to auto loan debt aversion**
- Credit card limits are **lower** but become **higher over time**

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Implication is that immigrants are profitable to lend to. Potential solutions?
- **FinTech innovations** (e.g., AmEx Global Card Relationship, NovaCredit, Global Credit File)
- **Removing barriers** to lending to immigrants or sharing information across borders
- **Shorten duration** for retaining positive information on credit reports, but this has trade-offs (e.g., Blattner, Nelson, & Hartwig, 2022; Kovbasyuk & Spagnolo, 2024)

Thank You!

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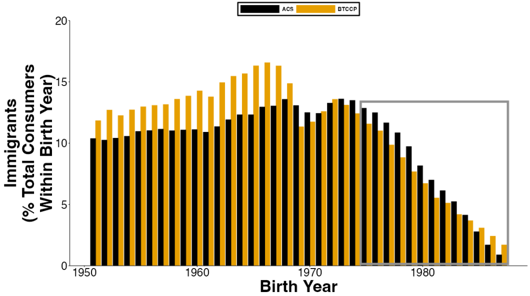
U.S. Consumer Credit Reporting Data: The BTCCP

- University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)
- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024. Includes:
  - Anonymized consumer identifiers
  - Individual tradeline data (e.g., details on each account, including opening dates)
  - Consumer-level data (e.g., birth date, credit score, ZIP code)
- See guidance in Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang (2025)

Our Immigrant Classification Aligns With American Community Survey (ACS)

Distribution of Immigrants by...

A. Birth Year



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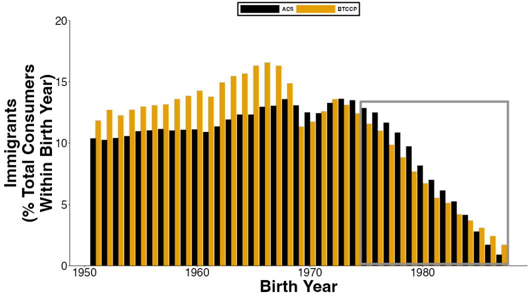
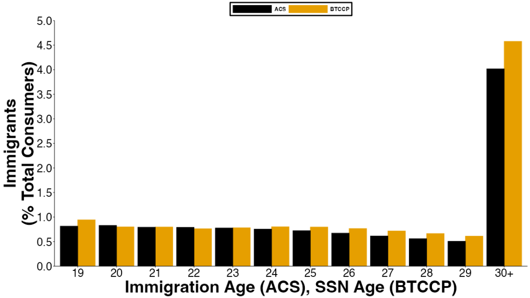


B. Immigration Age



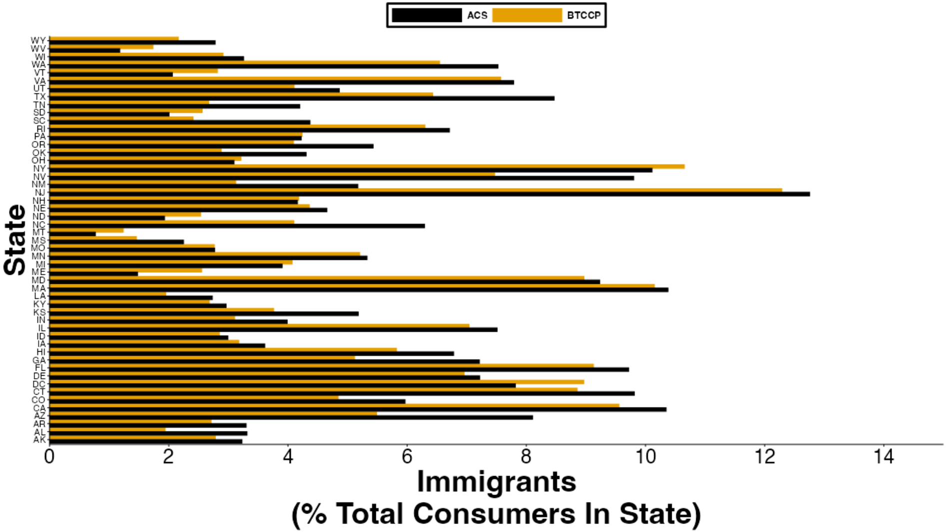
Distribution of immigrants also aligns by geography

Immigrants more geographically mobile than Non-Immigrants

"Entrant Sample": Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry In Ages 21 To 29 & Credit Outcomes For Ages 18 To 40

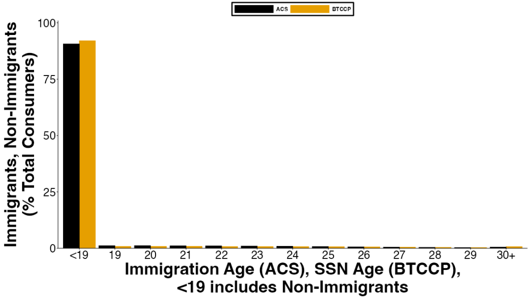
TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
Clean Sample	18,572,654
... SSN Age $<$21	16,478,940
... SSN Age 21+	2,093,714 (11.27%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
Entrant Sample (SSN Age $<$ 30)	6,122,932
... SSN Age $<$ 21	5,778,671
... SSN Age 21-29	344,261 (5.96%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

Geographic Comparison of Immigrants: ACS vs. BTCCP

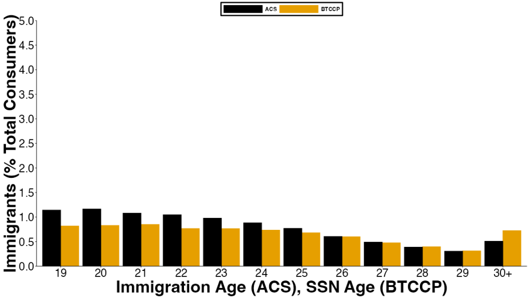


Immigrant Classification: ACS vs. BTCCP

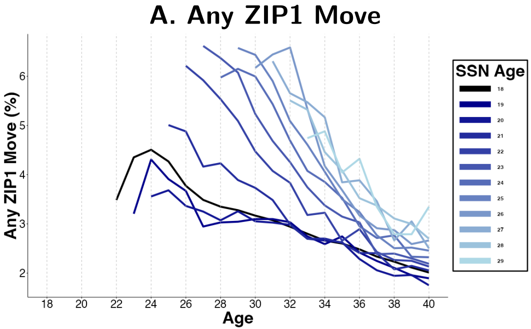
A. Non-Immigrants and Immigrants by Immigration Age

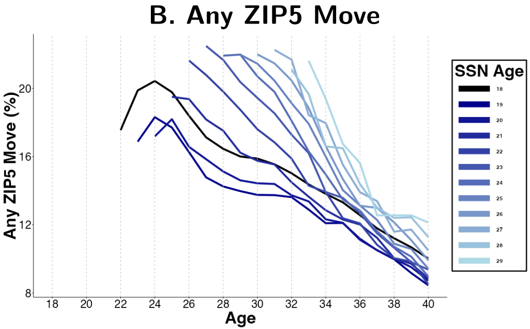


B. Immigrants by Immigration Age

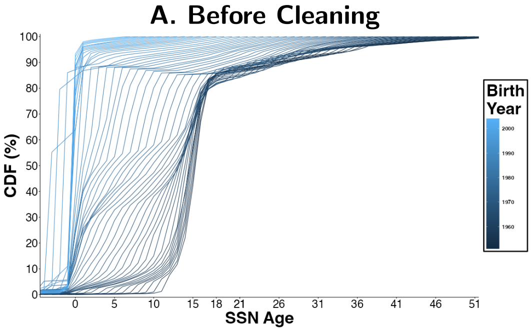


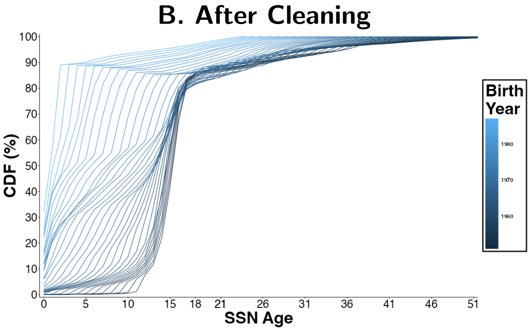
Geographic Mobility





SSN Ages By Birth Year





Consumers By SSN Age

SSN Age	Total	Entrants
$<$19	16,155,157	5,677,001
19	175,194	50,475
20	148,589	51,195
21	148,019	52,450
22	141,492	47,302
23	145,218	47,214
24	149,000	45,314
25	147,899	42,005
26	142,319	36,976
27	133,147	29,383
28	123,338	24,415
29	113,435	19,202
30+	849,847	44,642

Sample Frame and Timing of SSN Assignment

Calendar year	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987
	Birth year												
1993	≤18												
1994	19	≤18											
1995	20	19	≤18										
1996	21	20	19	≤18									
1997	22	21	20	19	≤18								
1998	23	22	21	20	19	≤18							
1999	24	23	22	21	20	19	≤18						
2000	25	24	23	22	21	20	19	≤18					
2001	26	25	24	23	22	21	20	19	≤18				
2002	27	26	25	24	23	22	21	20	19	≤18			
2003	28	27	26	25	24	23	22	21	20	19	≤18		
2004	29	28	27	26	25	24	23	22	21	20	19	≤18	
2005	30	29	28	27	26	25	24	23	22	21	20	19	≤18
2006	31	30	29	28	27	26	25	24	23	22	21	20	19
2007	32	31	30	29	28	27	26	25	24	23	22	21	20
2008	33	32	31	30	29	28	27	26	25	24	23	22	21
2009	34	33	32	31	30	29	28	27	26	25	24	23	22
2010	35	34	33	32	31	30	29	28	27	26	25	24	23
2011	36	35	34	33	32	31	30	29	28	27	26	25	24
2012	37	36	35	34	33	32	31	30	29	28	27	26	25
2013	38	37	36	35	34	33	32	31	30	29	28	27	26
2014	39	38	37	36	35	34	33	32	31	30	29	28	27
2015	40	39	38	37	36	35	34	33	32	31	30	29	28
2016	41	40	39	38	37	36	35	34	33	32	31	30	29
2017	42	41	40	39	38	37	36	35	34	33	32	31	30
2018	43	42	41	40	39	38	37	36	35	34	33	32	31
2019	44	43	42	41	40	39	38	37	36	35	34	33	32
2020	45	44	43	42	41	40	39	38	37	36	35	34	33
2021	46	45	44	43	42	41	40	39	38	37	36	35	34
2022	47	46	45	44	43	42	41	40	39	38	37	36	35
2023	48	47	46	45	44	43	42	41	40	39	38	37	36
2024	49	48	47	46	45	44	43	42	41	40	39	38	37

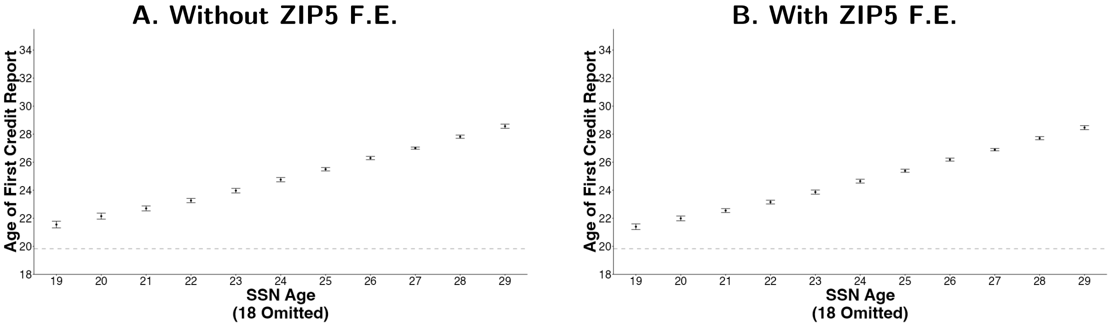
Age at SSN assignment

Years of data in our sample

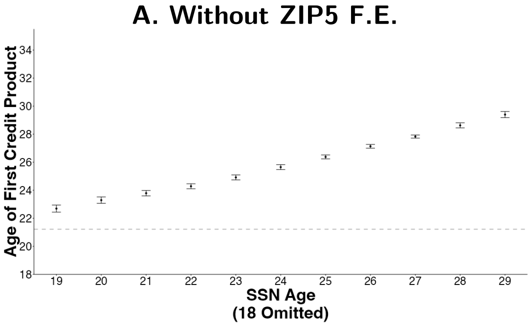
Summary Statistics

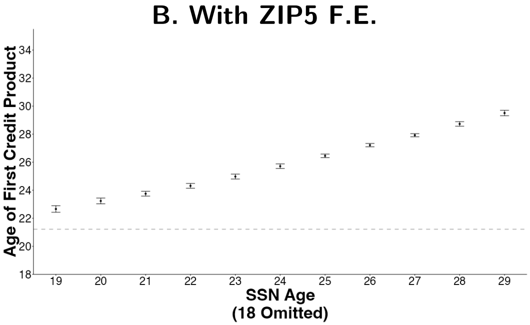
	Non-Immigrants *SSN Age ¡ 21*		Immigrants *SSN Age 21+*	
	Mean	# Consumers	Mean	# Consumers
Credit Report				
Age at First Credit Report	19.86	5,778,671	25.21	344,261
Credit Card				
Any Credit Card (%)	97.16	5,778,671	97.85	344,261
Age at First Credit Card	22.07	5,614,625	26.42	336,869
Credit Score				
Any Score by Age 30 (%)	97.57	5,778,671	88.14	344,261
Credit Score at Age 30	626.1	5,461,785	660.7	293,349
Any Score by Age 40 (%)	99.61	4,487,804	99.08	312,391
Credit Score at Age 40	659.5	4,179,752	698.9	270,177
Auto Loan				
Any Auto Loan (%)	82.11	5,778,671	69.07	344,261
Age at First Auto Loan	26.03	4,745,108	30.30	237,797
Mortgage				
Any Mortgage (%)	51.13	5,778,671	47.62	344,261
Age at First Mortgage	29.49	2,954,699	32.73	163,937
Credit Card Limits				
Age 30	11,733	5,778,671	11,193	344,261
Age 40	22,477	4,487,804	28,158	312,391

Age At First Credit Report

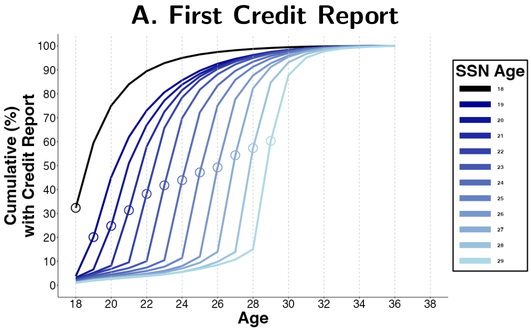


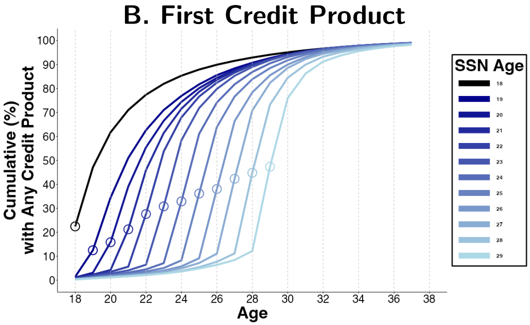
Age At First Credit Product





Age At First Credit Report and First Credit Product By SSN Age Cohort





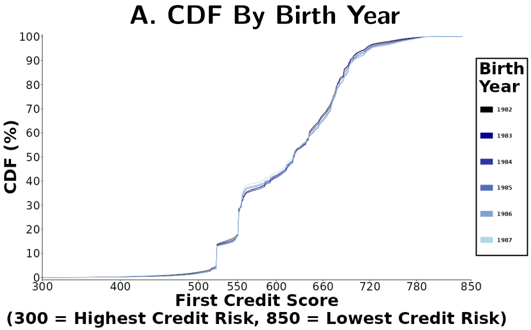
Credit Entry

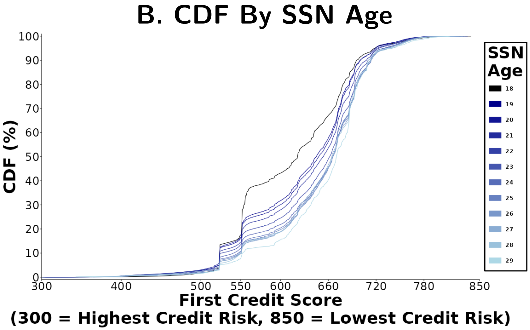
Dep Var: Age at First...	Credit Report			Credit Product		
	(1)	(2)	(3)	(4)	(5)	(6)
21+	1.989***	1.972***	1.809***	1.680***	1.665***	1.618***
	(0.119)	(0.101)	(0.088)	(0.129)	(0.112)	(0.106)
SSN Age	0.782***	0.740***	0.744***	0.744***	0.704***	0.720***
	(0.027)	(0.015)	(0.014)	(0.026)	(0.017)	(0.016)
Birth Year F.E.		X	X		X	X
First Zip5 F.E.			X			X
R^2	0.239	0.267	0.289	0.077	0.087	0.129
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	19.864	19.864	19.864	21.251	21.251	21.251

Credit Entry (Additional F.E.)

Dep Var: Age at First...		Credit Report			Credit Product	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	1.7757***	1.7721***	1.6915***	1.6322***	1.6353***	1.477***
	(0.0787)	(0.0779)	(0.0733)	(0.0907)	(0.0884)	(0.0878)
SSN Age	0.7466***	0.7463***	0.7273***	0.7271***	0.7264***	0.6886***
	(0.0133)	(0.0133)	(0.0144)	(0.0155)	(0.0154)	(0.0157)
Birth Year F.E.	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X
Number Zip5 F.E.			X			X
R^2	0.299	0.299	0.323	0.152	0.156	0.192
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	19.864	19.864	19.864	21.2508	21.2508	21.2508

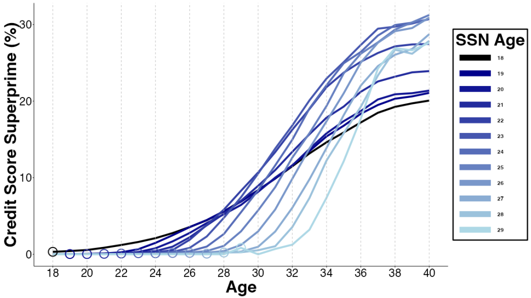
CDF Of First Credit Scores



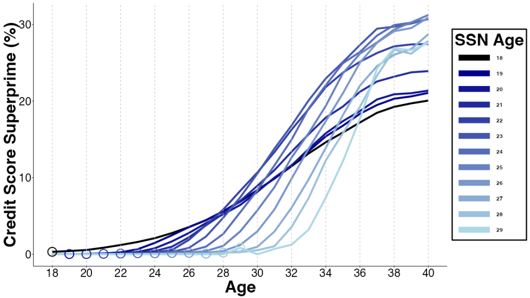


Lifecycle Of The Distribution Of Credit Scores By SSN Age Cohorts

A. Subprime (<600)



B. Superprime (780+)



Immigrant Credit Scores Higher In Spite Of Shorter Credit Histories

	Credit Score at		Any Prime or Higher (660+) Credit Score at	
	Age 30	Age 40	Age 30	Age 40
	(1)	(2)	(3)	(4)
21+	23.7***	32.3***	9.50***	4.08***
	(2.1)	(0.7)	(0.83)	(0.29)
SSN Age	3.2***	2.4***	0.29	1.92***
	(0.4)	(0.2)	(0.23)	(0.10)
Birth Year F.E.	X	X	X	X
R^2	0.009	0.022	0.006	0.0100
N	5,755,134	4,449,929	6,122,932	4,800,195
Mean, SSN Age <21	626.1	659.5	37.71	46.84

$$\text{Y}_i = \beta_1 \cdot \mathbb{I}\{SSN\ Age\ 21+\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \epsilon_i \tag{2}$$

Credit Score

Dep Var: Credit Score at ...		Age 30			Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	23.3***	23.7***	18.8***	31.5***	32.3***	25.8***
	(2.3)	(2.1)	(1.6)	(0.6)	(0.7)	(0.7)
SSN Age	2.7***	3.2***	2.0***	1.7***	2.4***	1.5***
	(0.5)	(0.4)	(0.4)	(0.4)	(0.2)	(0.2)
Birth Year F.E.		X	X		X	X
First Zip5 F.E.			X			X
R^2	0.005	0.009	0.135	0.008	0.022	0.132
N	5,755,134	5,755,134	5,755,134	4,449,929	4,449,929	4,449,929
Mean, SSN Age <21	626.1	626.1	626.1	659.5	659.5	659.5

Credit Score (Additional F.E.)

Dep Var: Credit Score at...		Age 30			Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	15.7***	15.4***	15.7***	21.6***	21.1***	21.6***
	(1.3)	(1.2)	(1.2)	(0.6)	(0.6)	(0.6)
SSN Age	1.5***	1.5***	1.6***	1.3***	1.3***	1.5***
	(0.3)	(0.3)	(0.3)	(0.1)	(0.1)	(0.2)
Birth Year F.E.	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X
Number Zip5 F.E.			X			X
R^2	0.193	0.201	0.202	0.192	0.199	0.200
N	5,755,134	5,755,134	5,755,134	4,449,929	4,449,929	4,449,929
Mean, SSN Age $<$21	626.1	626.1	626.1	659.5	659.5	659.5

Credit Score Prime or Higher (660+)

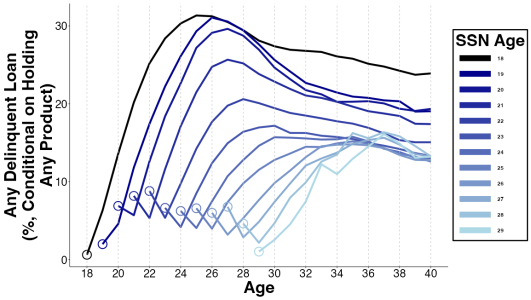
Dep Var: Prime or Higher at ...		Age 30			Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	9.35***	9.50***	7.63***	3.86***	4.08***	2.01***
	(0.93)	(0.83)	(0.74)	(0.31)	(0.29)	(0.14)
SSN Age	0.01	0.29	-0.08	1.71***	1.92***	1.58***
	(0.28)	(0.23)	(0.20)	(0.14)	(0.10)	(0.09)
Birth Year F.E.		X	X		X	X
First Zip5 F.E.			X			X
R^2	0.002	0.006	0.104	0.004	0.010	0.092
N	6,122,932	6,122,932	6,122,932	4,800,195	4,800,195	4,800,195
Mean, SSN Age $<$21	37.71	37.71	37.71	46.84	46.84	46.84

Credit Score Prime or Higher (660+), With Additional F.E.

Dep Var: Prime or Higher Score at...		Age 30			Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	6.45***	6.32***	6.60***	1.32***	1.19***	1.99***
	(0.60)	(0.55)	(0.57)	(0.23)	(0.27)	(0.21)
SSN Age	-0.22	-0.22	-0.14	1.43***	1.43***	1.58***
	(0.18)	(0.18)	(0.19)	(0.07)	(0.07)	(0.09)
Birth Year F.E.	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X
Number Zip5 F.E.			X			X
R^2	0.149	0.155	0.156	0.141	0.146	0.152
N	6,122,932	6,122,932	6,122,932	4,800,195	4,800,195	4,800,195
Mean, SSN Age $<$21	37.71	37.71	37.71	46.84	46.84	46.84

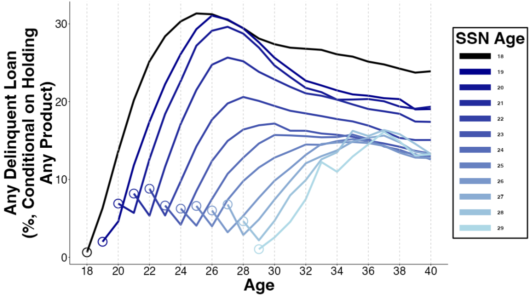
Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency

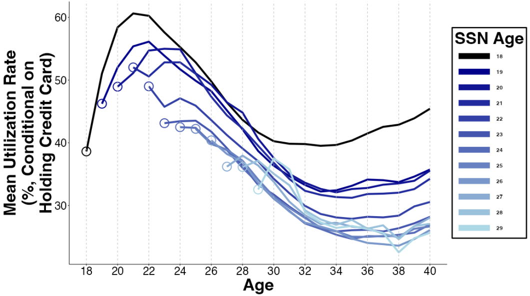


Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency

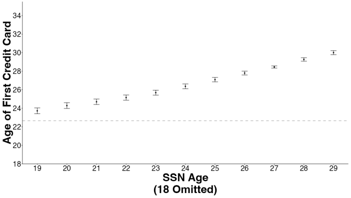


B. Credit Card Utilization Rate

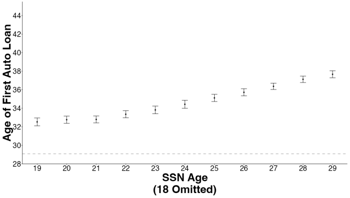


Age At First Credit Product By Type

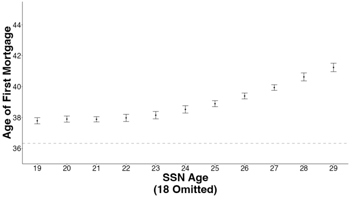
A. Credit Card



B. Auto Loan



C. Mortgage



Credit Access

Dep Var: Age at First...		Credit Card			Auto Loan			Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
21+	1.172***	1.160***	1.367***	2.954***	2.930***	1.967***	0.831***	0.781***	0.219*
	(0.182)	(0.169)	(0.158)	(0.239)	(0.217)	(0.217)	(0.124)	(0.097)	(0.083)
SSN Age	0.696***	0.663***	0.691***	0.673***	0.612***	0.619***	0.504***	0.403***	0.432***
	(0.030)	(0.022)	(0.020)	(0.045)	(0.023)	(0.024)	(0.048)	(0.018)	(0.018)
Birth Year F.E.		X	X		X	X		X	X
First Zip5 F.E.			X			X			X
R^2	0.029	0.033	0.085	0.025	0.030	0.082	0.008	0.024	0.081
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	22.677	22.677	22.677	29.151	29.151	29.151	36.356	36.356	36.356

Credit Access (Additional F.E.)

Dep Var: Age at First...		Credit Card			Auto Loan			Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
21+	1.4213***	1.4318***	1.1952***	1.7944***	1.7860***	1.4242***	0.0911	0.0735	-0.0822
	(0.1344)	(0.1308)	(0.1250)	(0.1736)	(0.1729)	(0.1356)	(0.0653)	(0.0671)	(0.0518)
SSN Age	0.7010***	0.7000***	0.6435***	0.6271***	0.6255***	0.5401***	0.4454***	0.4434***	0.4057***
	(0.0191)	(0.0190)	(0.0194)	(0.0208)	(0.0215)	(0.0230)	(0.0162)	(0.0163)	(0.0173)
Birth Year F.E.	X	X	X	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X		X	X
Number Zip5 F.E.			X			X			X
R^2	0.114	0.118	0.159	0.120	0.125	0.169	0.131	0.143	0.154
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	22.6771	22.6771	22.6771	29.1511	29.1511	29.1511	36.3559	36.3559	36.3559

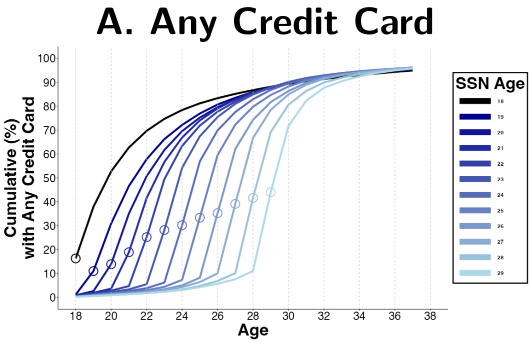
Credit Access By Age 37

Dep. Var: By Age 37, has ...		Credit Card			Auto Loan			Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
21+	0.93***	0.97***	0.15	-10.24***	-10.22***	-6.87***	-2.19***	-2.28***	0.39
	(0.19)	(0.16)	(0.16)	(0.38)	(0.37)	(0.46)	(0.55)	(0.56)	(0.53)
SSN Age	0.03	0.10*	0.04	-1.48***	-1.42***	-1.44***	-1.34***	-1.58***	-1.77***
	(0.05)	(0.04)	(0.03)	(0.10)	(0.09)	(0.10)	(0.1)	(0.09)	(0.09)
Birth Year F.E.		X	X		X	X		X	X
First Zip5 F.E.			X			X			X
R^2	$<$ 0.001	0.002	0.025	0.008	0.009	0.039	0.002	0.005	0.063
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age $<$21	94.93	94.93	94.93	77.22	77.22	77.22	45.46	45.46	45.46

Credit Access By Age 37 (Additional F.E.)

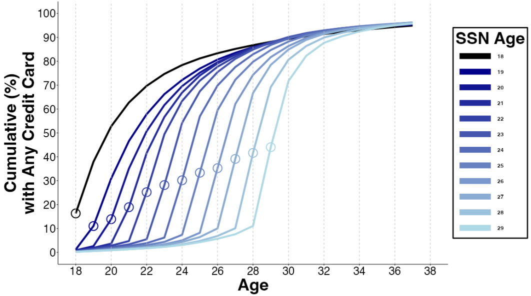
Dep Var: By Age 37, has...		Credit Card			Auto Loan			Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
21+	-0.02	-0.05	0.5***	-6.18***	-6.16***	-4.43***	1.09*	1.13*	2.33***
	(0.12)	(0.12)	(0.14)	(0.34)	(0.35)	(0.2)	(0.49)	(0.46)	(0.35)
SSN Age	0.02	0.02	0.15***	-1.48***	-1.47***	-1.06***	-1.86***	-1.85***	-1.56***
	(0.03)	(0.03)	(0.04)	(0.09)	(0.09)	(0.11)	(0.08)	(0.08)	(0.09)
Birth Year F.E.	X	X	X	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X		X	X
Number Zip5 F.E.			X			X			X
R^2	0.040	0.042	0.060	0.076	0.079	0.124	0.117	0.127	0.146
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	94.93	94.93	94.93	77.22	77.22	77.22	45.46	45.46	45.46

Immigrants’ Access to Credit Cards Converges With Non-Immigrants

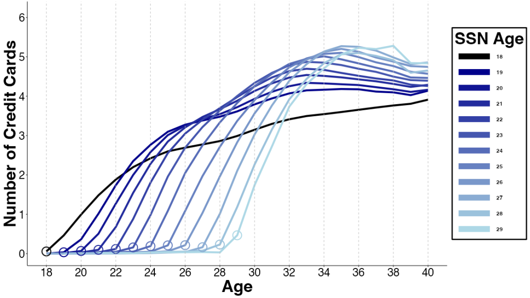


Immigrants' Access to Credit Cards Converges With Non-Immigrants

A. Any Credit Card



B. Number of Credit Cards

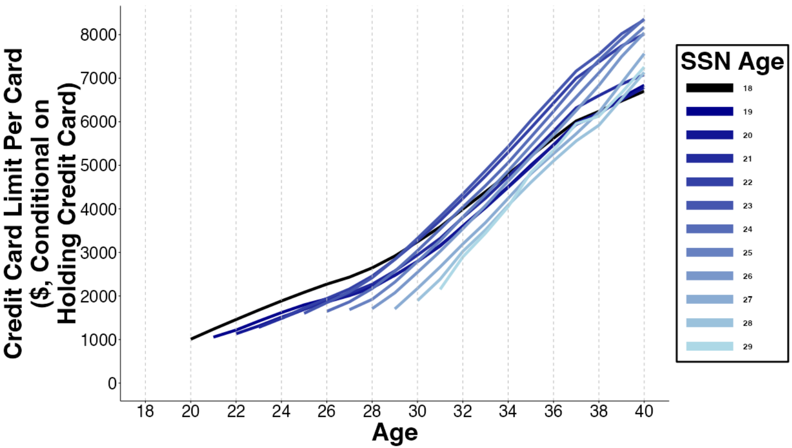


By Age 37, Immigrants are:

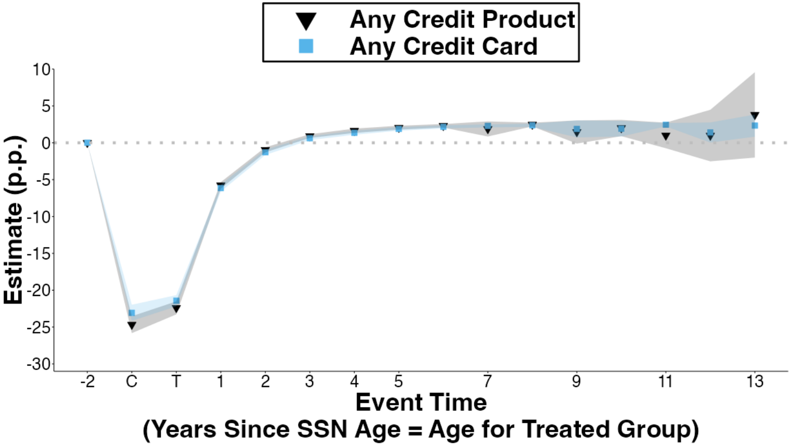
0.97 (s.e. 0.16) p.p. (1%) *more* likely to have accessed credit card

- No robust result for older SSN Ages affecting credit card use

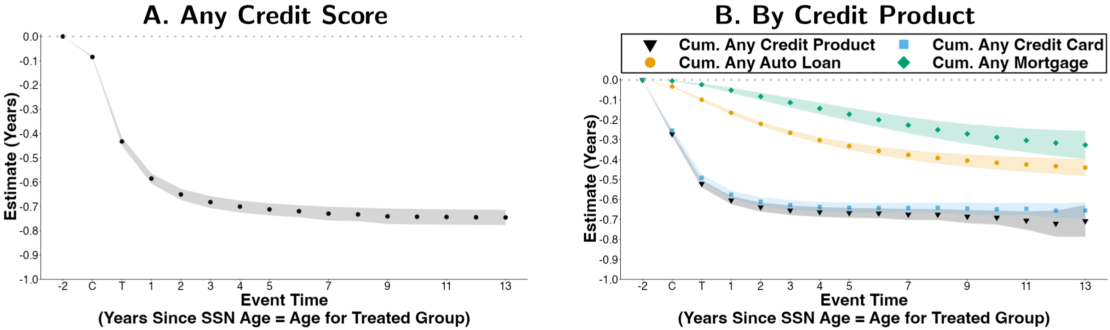
Credit Card Limit Per Card



Paired Cohorts: First Credit Product & First Credit Card



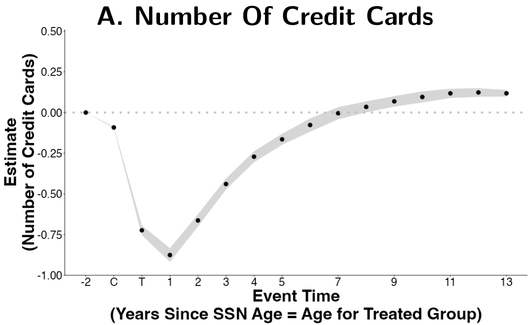
Paired Cohorts: Cumulatives

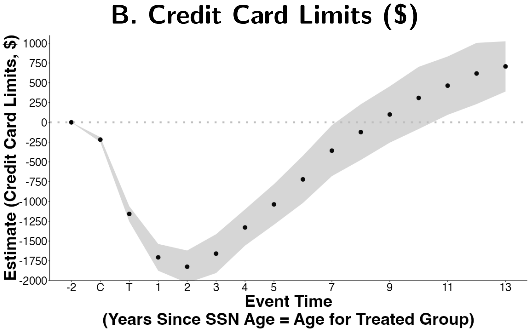


Paired Cohorts: Cumulative Years of Delay

	Any Credit Score	Credit Product	Credit Card	Auto Loan	Mortgage
	(1)	(2)	(3)	(4)	(5)
C	-0.085***	-0.272***	-0.254***	-0.033***	-0.005***
	(0.002)	(0.005)	(0.005)	(0.002)	(0.001)
T	-0.432***	-0.521***	-0.491***	-0.100***	-0.024***
	(0.009)	(0.009)	(0.009)	(0.004)	(0.003)
5	-0.712***	-0.667***	-0.640***	-0.332***	-0.173***
	(0.013)	(0.013)	(0.013)	(0.010)	(0.017)
10	-0.742***	-0.690***	-0.648***	-0.415***	-0.288***
	(0.016)	(0.018)	(0.016)	(0.017)	(0.028)
N Paired Cohorts	68	68	68	68	68
N Consumers	403,506	403,506	403,506	403,506	403,506
N	6,456,096	6,456,096	6,456,096	6,456,096	6,456,096

Paired Cohorts: Credit Cards





Taking Stock Of Our Key Results

We Find That Immigrants vs. Non-Immigrants,
& Immigrants Arriving At Older (vs. Younger) Ages Are:

- **More creditworthy** (once credit scored)
- **Delayed entry** into credit system
- **Lower long-term access** to auto loans **due to auto loan debt aversion**
- Credit card limits are **lower** but become **higher over time**

Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

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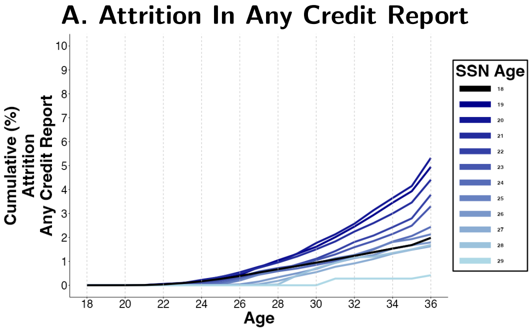
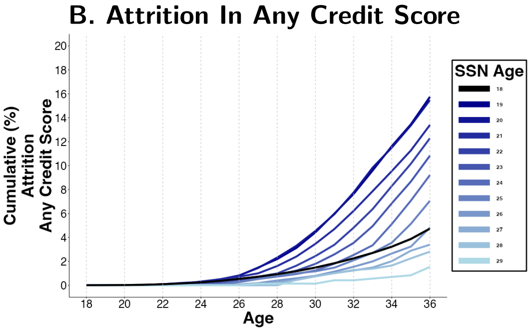
Restrict to consumers observed in 2024:

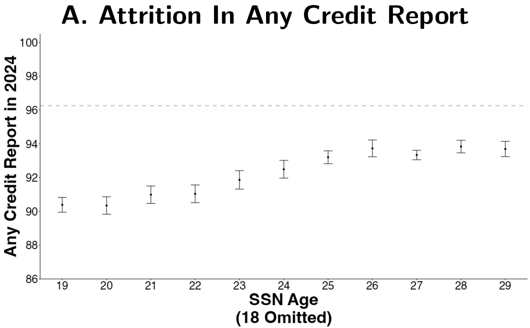
- **Mortgage** Immigrant vs. Non-Immigrant gap closes butSSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

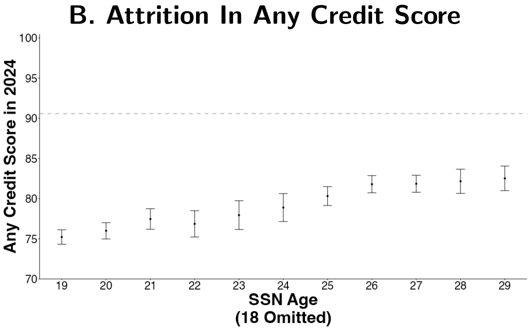
Attrition


A. Attrition In Any Credit Report
Cumulative (%) Attrition Any Credit Report
Age
SSN Age
18
19
20
21
22
23
24
25
26
27
28
29


B. Attrition In Any Credit Score
Cumulative (%) Attrition Any Credit Score
Age
SSN Age
18
19
20
21
22
23
24
25
26
27
28
29

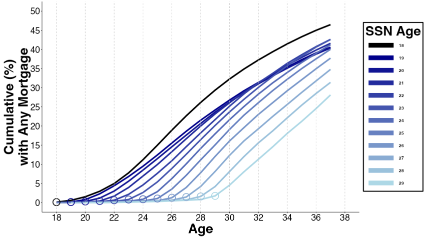
Consumers Observed In 2024



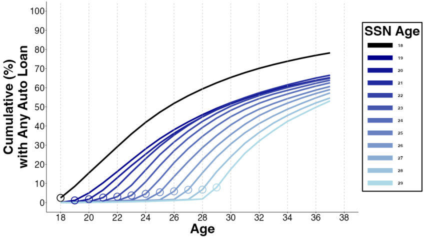


Lifecycle Of Credit For Consumers Observed In 2024

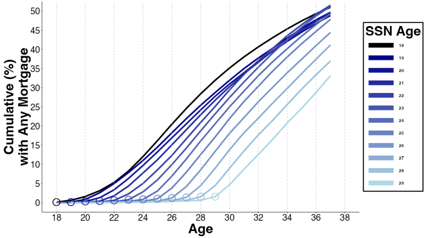
A. Mortgages (Any 2024 Report)



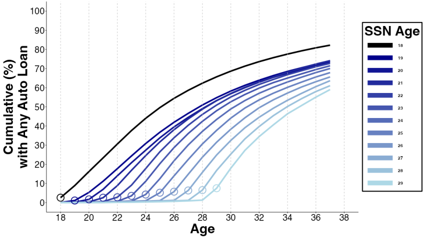
B. Auto Loan (Any 2024 Report)



C. Mortgages (Any 2024 Product)

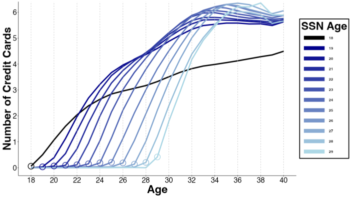


D. Auto Loan (Any 2024 Product)

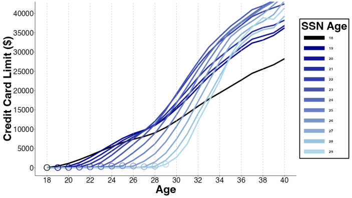


Lifecycle Of Credit For Consumers Observed In 2024 (Any Tradeline)

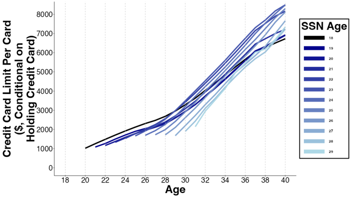
A. Number Credit Cards



B. Credit Card Limits



C. Credit Limit Per Card



Consumers Observed In Data In 2024 (Additional F.E.)

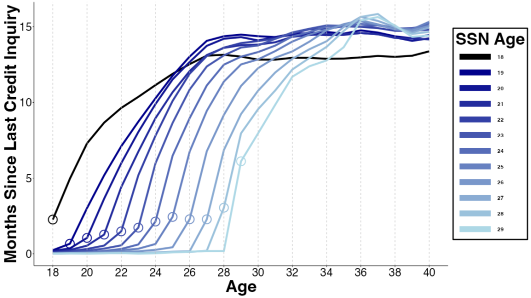
Dep Var: In 2024, has...	Any Credit Report		Any Credit Score		Any Credit Tradeline	
	(1)	(2)	(3)	(4)	(5)	(6)
21+	-0.0543***	-0.0485***	-0.1404***	-0.1216***	-0.1442***	-0.1207***
	(0.0020)	(0.0022)	(0.0070)	(0.0048)	(0.0063)	(0.0042)
SSN Age	0.0043***	0.0052***	0.0080***	0.0109***	0.0068***	0.0108***
	(0.0005)	(0.0004)	(0.0009)	(0.0007)	(0.0007)	(0.0006)
Birth Year F.E.	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X
Last Zip5 F.E.		X		X		X
Longest Zip5 F.E.		X		X		X
Number Zip5 F.E.		X		X		X
R^2	0.007	0.034	0.020	0.111	0.023	0.138
N	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932	6,122,932
Mean, SSN Age <21	0.9615	0.9615	0.9031	0.9031	0.8503	0.8503

Credit Access By Age 37 (Additional F.E.) Conditional On Any Credit Score Observed in 2024

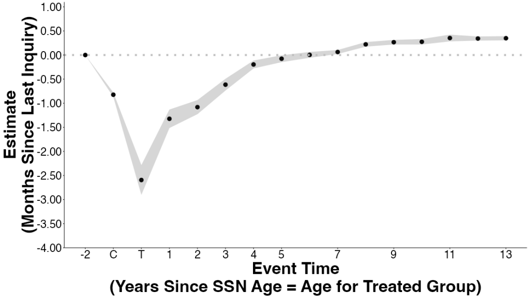
Dep Var: By Age 37, has...		Credit Card			Auto Loan			Mortgage	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
21+	0.56***	0.51***	0.67***	-2.56***	-2.56***	-2.01***	6.33***	6.31***	6.57***
	(0.11)	(0.11)	(0.13)	(0.34)	(0.35)	(0.22)	(0.49)	(0.48)	(0.41)
SSN Age	-0.04	-0.03	0.08	-1.83***	-1.83***	-1.46***	-2.45***	-2.44***	-2.20***
	(0.03)	(0.03)	(0.04)	(0.10)	(0.10)	(0.12)	(0.08)	(0.08)	(0.11)
Birth Year F.E.	X	X	X	X	X	X	X	X	X
First Zip5 F.E.	X	X	X	X	X	X	X	X	X
Last Zip5 F.E.	X	X	X	X	X	X	X	X	X
Longest Zip5 F.E.		X	X		X	X		X	X
Number Zip5 F.E.			X			X			X
R^2	0.038	0.042	0.054	0.067	0.071	0.099	0.114	0.125	0.134
N	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726	5,489,726
Mean, SSN Age <21	96.05	96.05	96.05	80.37	80.37	80.37	48.76	48.76	48.76

Potential Mechanism #3: Does Lower Credit Demand Explain Lower Access?
Immigrants Have Higher Credit Demand In 20s, Lower Demand In 30s

A. Months Since Last Credit Inquiry
Higher Value = Lower Credit Demand



B. Paired Cohorts
Months Since Last Credit Inquiry



Potential Mechanism #3: Does Lower Credit Demand Explain Lower Access? Immigrants Have Higher Credit Demand In 20s, Lower Demand In 30s

A. Months Since Last Credit Inquiry
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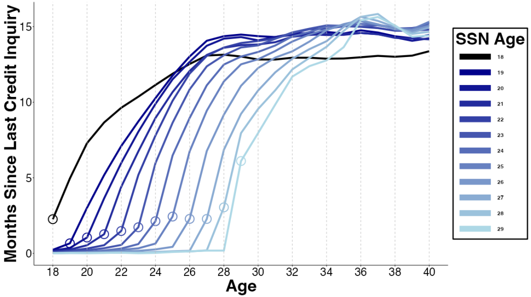
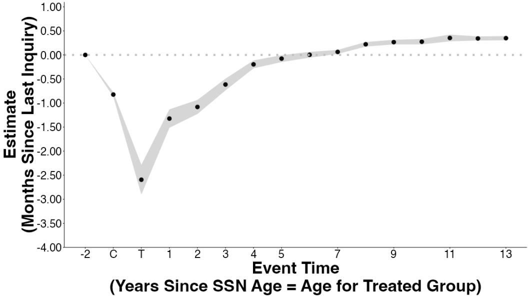


B. Paired Cohorts
Months Since Last Credit Inquiry



Caveat: Lower credit demand in 30s may be function of earlier rejections → deterrence

Potential Mechanism #3: Tastes

Immigrants and Non-Immigrants differences may be explained by tastes:

- Immigrants have different cultural backgrounds, may be less familiar/willing to use debt
- Autos & Property: Immigrants may choose not to purchase or purchase without debt

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Tastes unlikely to explain SSN Age gaps or paired cohorts results